

Healthcare Readmission & Hospital Performance Analysis

End-to-end EDA on **101,766 patient records** — uncovering readmission patterns, demographic trends, and hospital utilization insights.

PYTHON · PANDAS · EDA · KPI ANALYSIS

Dataset at a Glance

101K

Patient Records

50

Variables

71K

Unique Patients

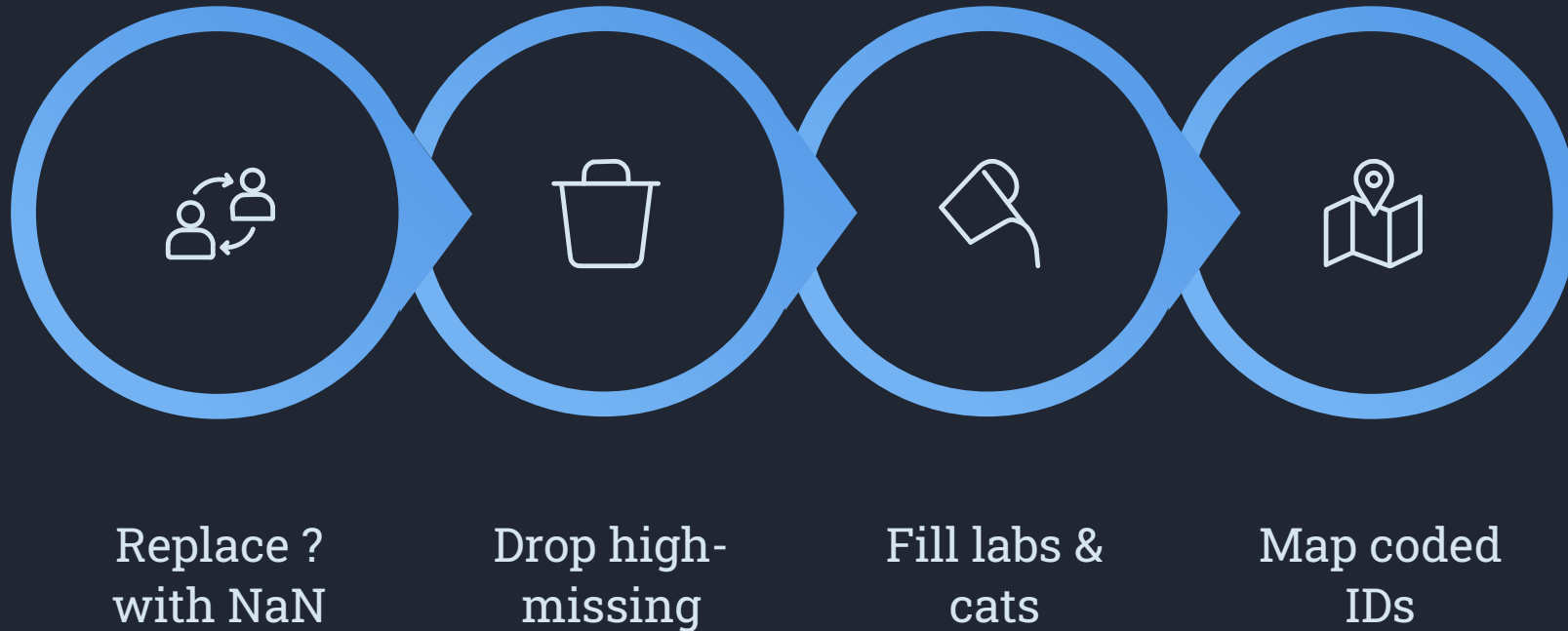
37

Categorical Cols

Data Quality Issues Found

- Placeholder "?" values in weight, payer_code, medical_specialty
- 94.75% missing in max_glu_serum, 83.28% in A1Cresult
- No duplicate records detected

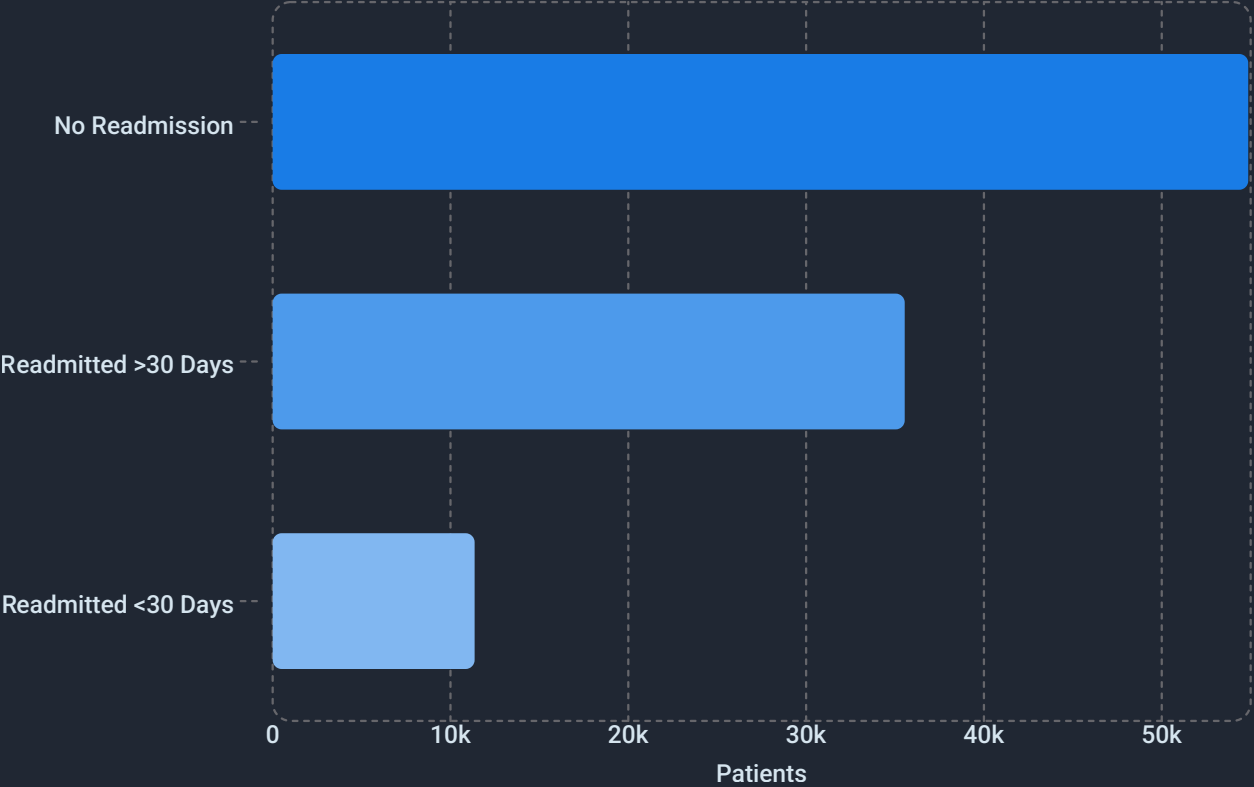
Data Cleaning Pipeline



Cleaned dataset: **101,766 rows × 51 columns** — zero missing values remaining.

Readmission Overview

Readmission Status



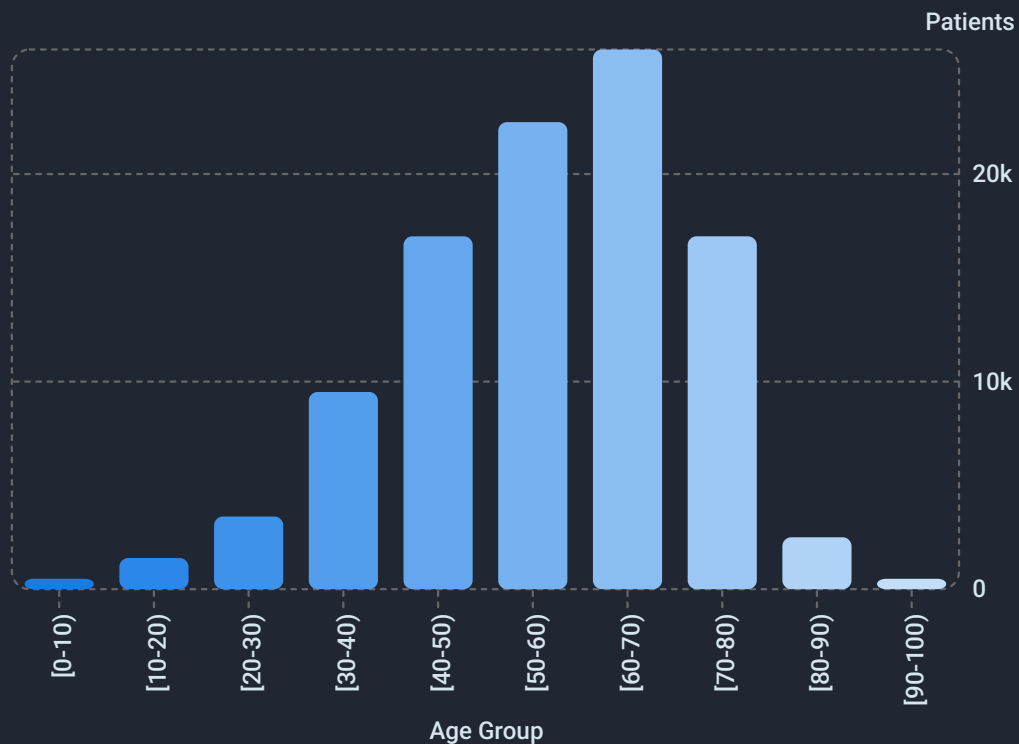
Key Findings

53.9%
No readmission

34.9%
Readmitted after 30 days

11.2%
Readmitted within 30 days

Patient Demographics



Demographic Highlights

- Peak age group: **60–80 years**
- Female patients slightly outnumber males (~54K vs ~47K)
- Caucasian patients dominate (~75K), followed by AfricanAmerican (~18K)

Older adults bear the highest diabetes-related hospitalization burden — target preventive care here.

Hospital & Treatment Metrics &

4.4 Days

Avg hospital stay

7.4

Avg diagnoses per patient

16.0

Avg medications prescribed

43.1

Avg lab procedures

Emergency admissions (~52K) and physician referrals (~28K) were top entry pathways. **78,363 patients** received diabetes medication; **47,011** had medication changes during stay.

Correlation Analysis

Correlation Heatmap



Correlation Insights

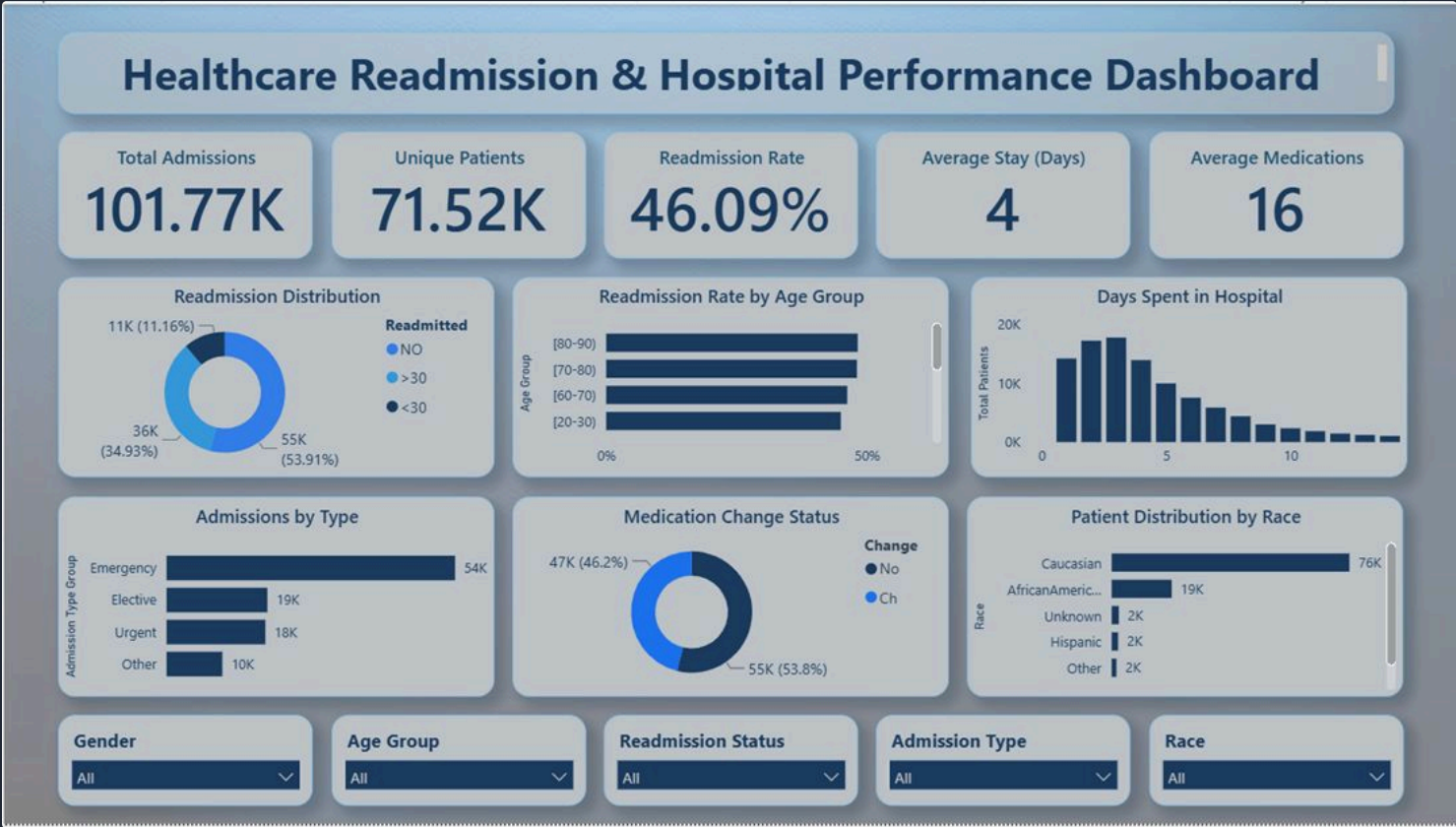
Medications vs
Hospital Stay: **0.47**

Lab Procedures vs
Hospital Stay: **0.32**

Medications vs
Procedures: **0.39**

Most variables show weak-to-moderate correlations, indicating that patient outcomes are influenced by multiple clinical factors rather than a single metric.

Healthcare Readmission & Hospital Performance Dashboard



Key Findings

- Interactive healthcare dashboard
- Readmission trend analysis
- Patient demographic insights
- Admission type comparison
- Dynamic slicers for data exploration